ISS 2103 • EXAM REVISION PACK 01

# Fundamentals: Components, Modes, Network Types & Topologies

Course outline weeks 1–2: concepts, goals & applications, network structure, topology design

Priority: **HIGH**

This topic appears in **every** paper, usually as 4–12 mark "explain / differentiate / draw" parts. These are the easiest marks on the paper: learn the definitions, the tables and the diagrams in this pack.

## How often this topic is examined

| Question asked | Paper(s) | Marks |
|---|---|---|
| Network topologies (describe 3 or 4) | 2025 Q2c, ISA2024 Q2c | 9–12 |
| LAN / MAN / WAN, categories of networks | ISA2024 Q3b, 2018 Q1c | 6–9 |
| Client-server vs peer-to-peer (incl. benefits, HIT example) | 2025 Q1b, SE2024 Q4b, Q5d | 4–8 |
| Components of a data communication system (diagram) | 2018 Q1a, SE2024 Q2f | 5 |
| What is a protocol (with examples) | 2025 Q2a | 5 |
| Transmission modes; half duplex; point-to-point | 2023 Q4e, SE2024 Q1a.i, Q1a.ii | 1–6 |
| Network design: 5 floors × 20 PCs | SE2024 Q3e | 5 |
| Configurations by relationship among nodes | SE2024 Q3d | 4 |
| Hardware, software & protocols of a network | SE2024 Q4d | 4 |
| Internet vs World Wide Web | SE2024 Q4a | 4 |
| Ring vs bus limitations; logical vs physical topology | SE2024 Q2e, Q3c.i | 2–3 |
| Network administrator / engineer competencies | SE2024 Q4e.i, Q4e.ii | 1–2 |

**EXAM TIP**

The ISA paper says "illustrate your answer, where appropriate, with large clearly labelled diagrams". **Draw every topology you describe** and every comparison you can. A small sketch is quick and earns marks.

## 1 Data communication: components and characteristics

**Data communication** is the exchange of data between two devices over a transmission medium. It needs **hardware** (sender, receiver, intermediate devices) and **software** (the rules, i.e. the **protocol**).




Figure: the five components of a data communication system. M-S-R-T-P.

**The four characteristics of an effective data communication system**

**Delivery**: data reaches the correct destination and user. **Accuracy**: delivered without errors. **Timeliness**: delivered on time (audio/video need real-time delivery). **Jitter**: the variation in packet arrival time; uneven jitter spoils real-time audio/video.

**PAST PAPER** **2018 Q1a / SE2024 Q2f** **[5 marks]**

With the aid of a fully labelled diagram, explain all the components involved in a data communication system. / With the aid of diagram, describe the complete computer communication systems.

Lecturer's 4 characteristics: **D-A-T-J**, Delivery, Accuracy, Timeliness, Jitter. Do not confuse them with the 5 components.

**Model answer**

Draw the figure above, then one line per component:

1. **Message**: the information to be communicated: text, numbers, images, audio or video.
2. **Sender**: the device that sends the message, e.g. a computer, phone or video camera.
3. **Receiver**: the device the message is meant for, e.g. a computer, printer or TV.
4. **Transmission medium**: the physical path from sender to receiver; wired (twisted pair, coaxial, fibre optic) or wireless (radio, microwave, infrared).
5. **Protocol**: an agreed set of rules that governs the communication. Without it two connected devices cannot understand each other, like two people speaking different languages.

**PAST PAPER** **SE2024 Q4d** **[4 marks]**

A basic computer network must consist of at least hardware, software and protocols. Give examples of each of these networking requirements and explain their implementation.

"Complete computer communication system" (SE2024) = the same 5-component diagram. You may add a modem/NIC or switch between sender and medium to show the hardware path.

**Model answer**

| Requirement | Examples | Implementation / role |
|---|---|---|
| **Hardware** | PCs, servers, NICs, UTP and fibre cables, switches, routers, wireless access points | Physically connect devices and carry signals; a switch joins devices in a LAN, a router joins networks |
| **Software** | Network OS (Windows Server, Linux), NIC drivers, browsers, email clients, firewall software | Manages users, shared resources and the applications that use the network |
| **Protocols** | TCP/IP, Ethernet, HTTP, DNS, DHCP | Rules so devices understand each other: addressing, delivery, error control |

## 2 Data flow: transmission modes and line configuration

The **transmission mode** is the direction of signal flow between two connected devices.

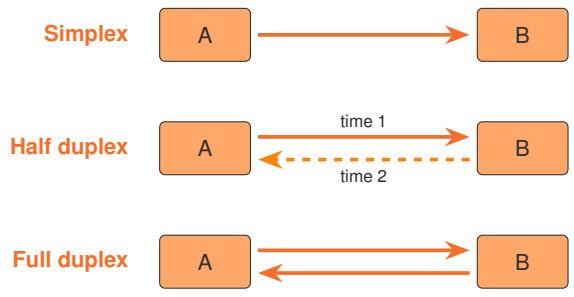



Figure: the three transmission modes (data flow).

**PAST PAPER** **2023 Q4e** **[6 marks]**

The transmission mode defines the direction of signal flow between two connected devices. State the three (3) transmission modes, giving examples for each mode.

**Model answer**

1. **Simplex**: communication is unidirectional; one device only sends, the other only receives. Examples: keyboard to computer, computer to monitor, radio/TV broadcast.
2. **Half duplex**: both stations can send and receive, but not at the same time; when one sends the other can only receive. Examples: walkie-talkie, CB radio.
3. **Full duplex**: both stations send and receive simultaneously. Examples: telephone or mobile phone call, video call.

**PAST PAPER** **SE2024 Q1a.i** **[1 marks]**

Explain the following term: Half Duplex Transmission.

**Model answer**

Two-way transmission in which data flows in both directions but **only one direction at a time**, e.g. a walkie-talkie (press to talk, release to listen).

## 2.1 Line configuration: point-to-point vs multipoint




***point-to-point**: whole link for 2 devices* *multipoint (multidrop): 3+ devices share one link*

Figure: the two types of connection (line configuration).

**PAST PAPER** **SE2024 Q1a.ii** **[1 marks]**

Explain the following term: Point to Point communication.

**Model answer**

A **dedicated link between exactly two devices**; the whole capacity of the link is reserved for them, e.g. a computer cabled directly to a printer, or a leased line between two routers. (Opposite: multipoint, where three or more devices share one link.)

Synchronous vs asynchronous transmission (SE2024 Q1a.iii) is in Pack 03.

# 3 Categories of networks (by size)

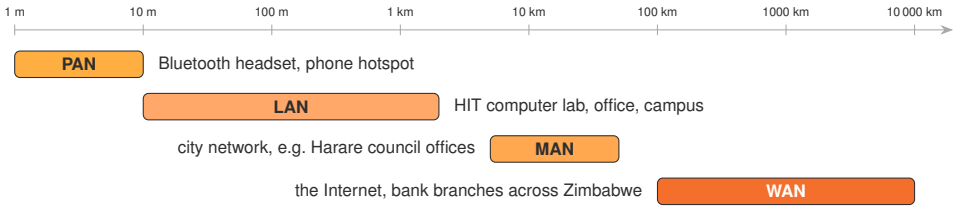



Figure: network categories by geographic span (log scale).

**PAST PAPER** **ISA2024 Q3b / 2018 Q1c** **[9 / 6 marks]**

Describe the differences between LAN, WAN, and MAN. / There are various categories of networks. Explain any three (3).

**Model answer**

| Feature | LAN (Local Area Network) | MAN (Metropolitan Area Network) | WAN (Wide Area Network) |
|---|---|---|---|
| **Coverage** | Room, building or campus (up to a few km) | A city or town (about 5–50 km); can join several LANs | Country, continent, world; joins LANs and MANs |
| **Ownership** | Privately owned by one organisation | Often a private provider, ISP or municipality | Public carriers / telcos (TelOne, Liquid), leased lines |
| **Speed** | Very high (100 Mbps – 10 Gbps) | High | Lower; depends on the link bought |
| **Errors, delay** | Low error rate, low delay | Moderate | Higher error rate and propagation delay |
| **Cost** | Cheap, easy to set up and maintain | Expensive to implement and maintain | Very expensive; cost varies with configuration |
| **Technology** | Ethernet, Wi-Fi; switches | Fibre rings, metro Ethernet, WiMAX, cable TV | Routers, leased lines, MPLS, satellite, undersea cable |
| **Example** | HIT computer lab | Network linking a city's offices | The Internet; a bank's branch network |

For 2018 Q1c (any three, 6 marks): define each with size + ownership + one example (2 marks each). Others you may use: **PAN** (about 10 m, Bluetooth), **CAN** (campus), **WLAN** (wireless LAN), **SAN** (storage).

**PAST PAPER** **SE2024 Q4a** **[4 marks]**

Distinguish between the internet and the World Wide Web.

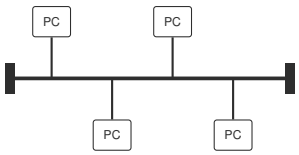
"Describe the differences" = a **table**. 9 marks: aim for 7 rows × 3 networks.

**Model answer**

| Internet | World Wide Web (WWW) |
|---|---|
| The global network of interconnected networks: the hardware and infrastructure | A service (application) that runs **on** the Internet |
| Uses TCP/IP to move packets | Uses HTTP/HTTPS, HTML and URLs |
| Carries many services: email, FTP, VoIP, gaming and the Web | A collection of linked web pages (hypertext documents) |
| Began 1969 as ARPANET | Invented by Tim Berners-Lee, 1989–1991 |

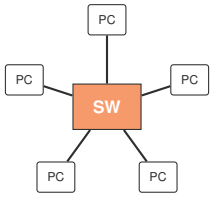
## 4 Network topologies

A **topology** is the layout pattern of the interconnections of nodes and links. **Physical topology** = the actual layout of cables and devices (how it looks). **Logical topology** = how data actually flows between devices (how it works).



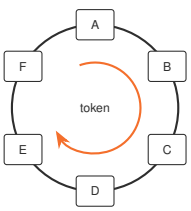

**Bus**

one backbone, terminator at each end



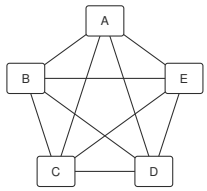

**Star**

every node to a central hub/switch



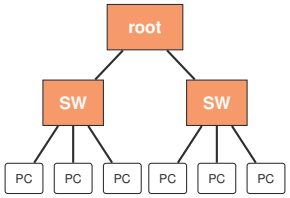

**Ring**

closed loop, data one way, token



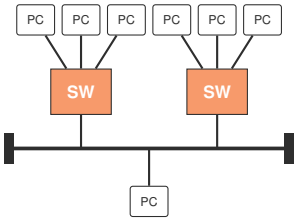

**Mesh (full)**

$n(n-1)/2$ links: 5 nodes = 10




**Tree (hierarchical)**

stars joined to a root/backbone




**Hybrid (star-bus)**

two or more topologies combined

Figure: the six physical topologies.

**PAST PAPER** **2025 Q2c / ISA2024 Q2c** **[12 / 9 marks]**

Describe any four (4) network topologies that can be used to connect devices. / Identify and explain any three (3) types of network topologies.

**Model answer**

For each topology: sketch + how it works + one advantage + one disadvantage (3 marks each).

| Topology | How it works | Advantages | Disadvantages |
|---|---|---|---|
| **Bus** | All devices attach by drop cables to one backbone cable, terminated at both ends. A signal travels both ways and every node checks the address | Cheap; least cable; easy to add a node | A break in the backbone stops the whole network; collisions; hard to troubleshoot; slows as nodes are added |
| **Star** | Each device has its own point-to-point cable to a central hub or switch; all traffic passes through it | Easy to install and wire; a cable fault affects one node only; easy fault isolation; add/remove without disruption | Hub/switch is a single point of failure; more cable and cost than bus |
| **Ring** | Each device connects to its two neighbours, forming a closed loop; data travels one way and each node repeats it (Token Ring, FDDI) | No collisions (token passing); equal access; performs well under heavy load | One break or one failed node can stop the ring; adding a node disrupts it; delay grows with nodes |
| **Mesh** | Every device has a dedicated point-to-point link to every other: $n(n-1)/2$ links, $n-1$ ports each | Very reliable, many paths; no shared-link traffic problems; private and secure | Very expensive; lots of cable and I/O ports; hard to install |
| **Tree** | A hierarchy: a root node links to second-level nodes, which link to the next level (stars joined to a backbone) | Scalable; easy to manage in sections; point-to-point wiring per segment | If the root/backbone fails, segments are cut off; more difficult to configure and wire |
| **Hybrid** | Two or more topologies combined, e.g. star-bus, star-ring | Flexible; uses the strengths of each; joins different LANs | Complex to design and manage; expensive |

**REMEMBER IT**

**Mesh maths**: links = $\frac{n(n-1)}{2}$, ports per device = $n-1$. 5 nodes $\rightarrow$ 10 links, 4 ports each. Choosing a topology: **cost, cable length, future growth, cable type, fault tolerance**.

**PAST PAPER** **SE2024 Q2e** **[3 marks]**

Explain the limitations of a ring network topology compared to a bus network topology.

**Model answer**

1. **Single break / failed node stops the ring**: data must pass through every node, so one failure breaks the loop (unless a dual ring is used). On a bus a failed node does not affect the others.
2. **Adding or removing a device disrupts the network**: the ring must be opened. On a bus you simply tap into the backbone.
3. **More delay and complexity**: every node must receive and regenerate the data, so delay grows with the number of nodes, and token management makes it costlier than a bus.

**PAST PAPER** **SE2024 Q3c.i** **[2 marks]**

Distinguish between the following terms: Logical verses Physical topology.

**Model answer**

**Physical topology** is the actual layout of the cables, devices and their locations (how the network looks). **Logical topology** is the way data flows between devices (how the network works). Example: Ethernet through a hub is a **physical star** but a **logical bus**; Token Ring wired to a MAU is a physical star but a logical ring.

The hub example scores the second mark: the same network can have different physical and logical topologies.

## 5 Network structure: backbone and local-access design

Large networks are designed in two levels:

- **Backbone network**: the high-capacity core that joins the sites (or floors, or buildings). Few nodes (routers, core switches), fast links (fibre, leased lines), usually a **mesh or partial mesh** so there are alternative paths.
- **Local-access network**: connects the end users' terminals to the nearest backbone node, usually a **star or tree** through switches, concentrators or multiplexers. Many cheap, lower-speed links.

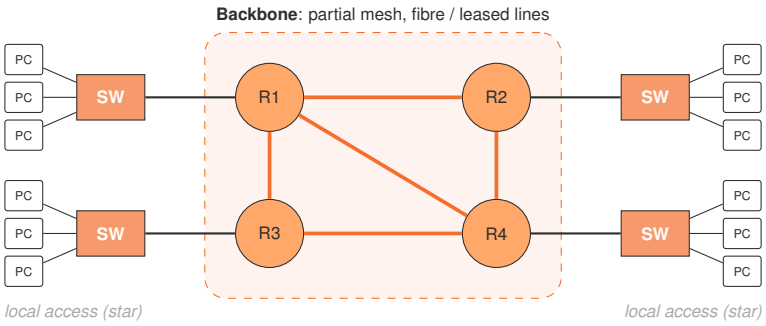



Figure: a two-level design. Users reach their local switch; sites talk over the backbone.

**PAST PAPER** **SE2024 Q3e** **[5 marks]**

Identify appropriate network topology and network connecting devices for following requirements. Draw a network design for proposed network for the following scenario. An organization having its office in a building of 5 floors. Each floor needs 20 machines.

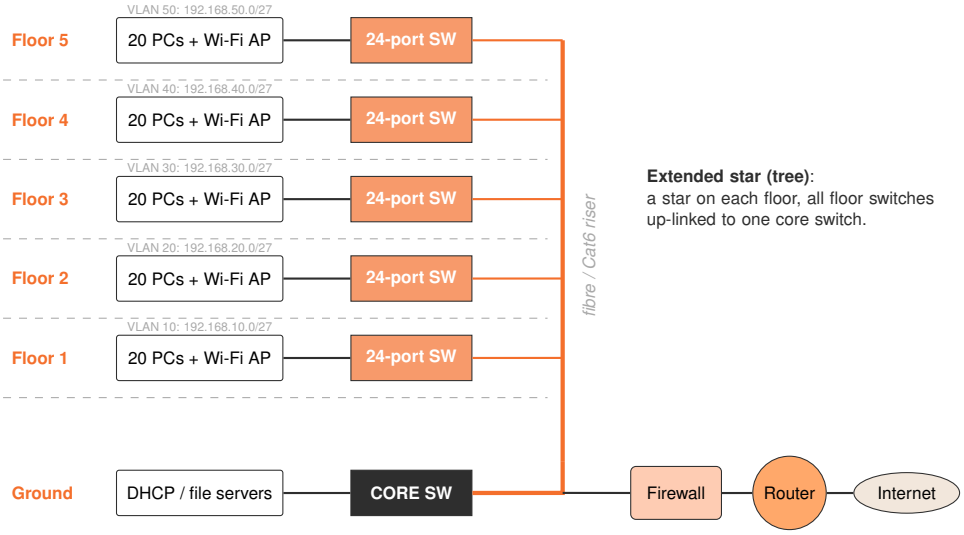
**Model answer**

- **Topology**: extended star (hierarchical/tree). A star on each floor; each floor switch up-links to a core switch over the vertical backbone (riser).
- **Devices**: a 24-port access switch per floor; a core switch; a router to the ISP; a firewall; wireless access points; Cat6 UTP to desks and fibre for the riser; servers in the ground-floor server room.

- **Addressing**: one VLAN/subnet per floor, addresses by DHCP.
- **Why**: easy to manage and expand; a failed floor switch only affects that floor.

**Sizing**

$$\text{machines} = 5 \times 20 = 100$$
$$\text{ports per floor} = 20 + 1\ (\text{AP}) + 1\ (\text{uplink}) = 22 \leq 24 \quad \leftarrow \textit{so a 24-port switch fits, 2 spare}$$
$$/27 \Rightarrow 2^5 - 2 = 30 \text{ hosts per floor} \quad \leftarrow \textit{20 PCs + growth}$$




**Extended star (tree)**:
a star on each floor, all floor switches up-linked to one core switch.

Figure: network design for a 5-floor building, 20 machines per floor.

## 6 Relationships among nodes: client-server and peer-to-peer

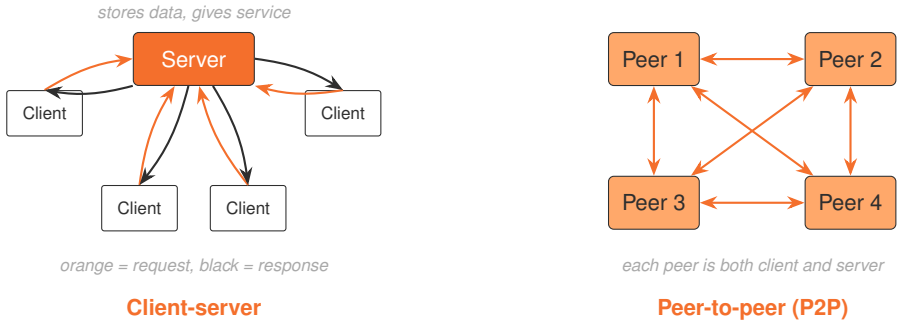



Figure: the two relationship models.

**PAST PAPER** **2025 Q1b** **[8 marks]**

With the aid of proper examples, differentiate between the Client-Server model and the Peer-to-Peer model for network applications.

**Model answer**

In the **client-server** model dedicated servers provide services that clients request (the web: a browser asks a web server for a page).
In the **peer-to-peer** model every node is equal and both requests and provides resources (BitTorrent: each peer downloads pieces from, and uploads pieces to, other peers).

| Point | Client-server | Peer-to-peer (P2P) |
|---|---|---|
| **Structure** | Dedicated, powerful server(s) provide services; clients request them | All nodes are equal peers; each acts as client and server |
| **Control & security** | Centralised: user accounts, access control on the server | Each user manages own machine; weaker security |
| **Data storage** | Stored centrally, easy backup | Scattered across peers; backup difficult |
| **Scalability** | Scales by adding/upgrading servers; server can become a bottleneck | Small office: slows as peers grow; file-sharing P2P scales because each peer adds capacity |
| **Cost** | Expensive: server hardware, network OS, administrators | Cheap: no dedicated server |
| **Failure** | Server is a single point of failure | No single point of failure |
| **Examples** | Web (browser ↔ web server), email (Outlook ↔ mail server), HIT e-learning portal, online banking | BitTorrent, Windows workgroup file sharing, blockchain/Bitcoin, early Skype |

**PAST PAPER** **SE2024 Q4b** **[4 marks]**

State the four benefits of client-server networks compared to peer to peer networks.

The question says "with the aid of proper examples". Name real applications on both sides or you cap your mark.

**Model answer**

1. **Centralised security**: one place for user authentication and access control.
2. **Centralised backup and data integrity**: all data on the server, backed up in one go.
3. **Easier management**: software updates, resources and users managed from the server.
4. **Better scalability and performance**: dedicated servers handle many clients; add servers as demand grows.

**PAST PAPER** **SE2024 Q3d** **[4 marks]**

Describe in detail four possible configurations of computer networks as categorized by relationships among nodes.

**Model answer**

1. **Peer-to-peer**: all computers are equal; each shares its own resources and acts as client and server. No central server.
2. **Client-server**: dedicated servers (file, mail, web, database) provide services to clients that request them.
3. **Point-to-point**: a dedicated link joins exactly two nodes.
4. **Multipoint (broadcast)**: many nodes share one link or channel; what one sends is received by all (bus, Wi-Fi).

**PAST PAPER** **SE2024 Q5d** **[7 marks]**

With reference to the client-server paradigm, explain how Harare Institute of Technology students are able to access online learning resources remotely from home.

The wording is ambiguous. Giving P2P, client-server, point-to-point and multipoint covers both readings of "relationship".

**Model answer**

1. The student's laptop or phone runs a browser: this is the **client**. It joins the Internet through the home ISP (Wi-Fi or mobile data).
2. The student enters the portal URL; the client asks a **DNS server** to resolve the name to the

server's IP address.
3. The client opens a **TCP connection** (3-way handshake) to the web server on port 443; **TLS** encrypts the session.
4. The client sends an **HTTP request**; the student logs in and the server **authenticates** the credentials against its user database.
5. The **server** processes the request, fetches notes, videos and assignments from central storage and sends HTTP **responses**.
6. The browser displays the resources; uploads (assignments) go the other way with HTTP POST.
7. One server serves **many clients at once**; resources are central, so every student sees the same updated material; firewalls/VPN protect the campus side.

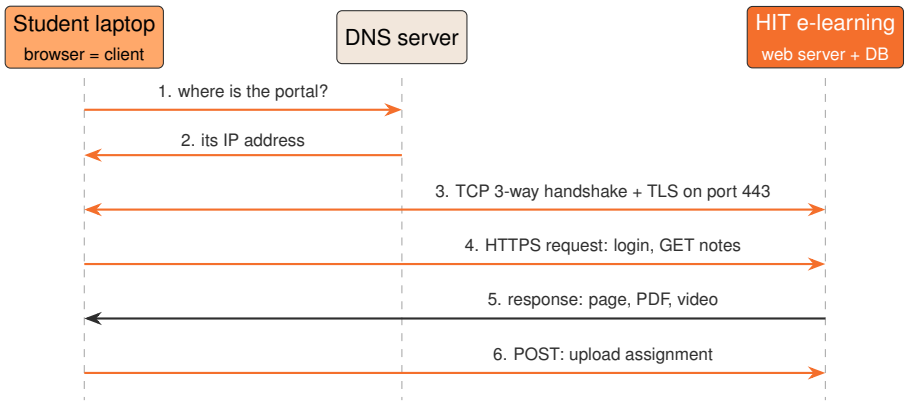



Figure: a remote student session in the client-server model (ISP and Internet in between).

## 7 Protocols and standards

A **protocol** is a set of rules that governs data communication: **what** is communicated, **how** and **when**.

**Three key elements of a protocol**

**Syntax**: the structure or format of the data (e.g. first 8 bits = source address).
**Semantics**: the meaning of each section of bits and the action to take.
**Timing**: when data should be sent and how fast, so the receiver is not overwhelmed.

**Standards** ensure interconnectivity and interoperability, so products from different vendors work together. **De facto** = "by fact / convention", adopted by widespread use without approval (e.g. TCP/IP grew this way). **De jure** = "by law / regulation", approved by an officially recognised body. Bodies: **ISO**, **ITU-T**, **ANSI**, **IEEE** (802.3 Ethernet, 802.11 Wi-Fi), **EIA**, **IETF** (RFCs).

**PAST PAPER** **2025 Q2a** **[5 marks]**

Protocol is usually associated with the software components of the network, explain with the aid of examples what a protocol is?

**Model answer**

A **protocol** is an agreed set of rules that governs data communication between devices: it defines what is communicated, how and when. Without a common protocol, two connected devices are like two people speaking different languages. Key elements: **syntax** (format of the data), **semantics** (meaning of the bits and the action to take), **timing** (when and how fast to send). Examples:

- **HTTP**/**HTTPS**: request and deliver web pages.
- **SMTP** (send), **POP3**/**IMAP** (receive): email.
- **TCP** / **UDP**: reliable or fast transport; **IP**: addressing and routing.
- **DNS**: name to IP address; **DHCP**: automatic IP configuration; **FTP**: file transfer.

**REMEMBER IT**

Protocol elements: **S-S-T**, Syntax (format), Semantics (meaning), Timing (when / how fast).

## 8 Network professionals

**PAST PAPER** SE2024 Q4e.i / Q4e.ii **[1 + 2 marks]**

Write brief notes about the competencies expected of the following professionals: (i) A network administrator. (ii) A network Engineer.

**Model answer**

**(i) Network administrator** (day-to-day running): configures and monitors the network, manages user accounts and permissions, backups, patching and troubleshooting, keeps the network available.
**(ii) Network engineer** (design and build): plans and designs networks (topology, addressing, routing); selects, installs and configures routers, switches and firewalls; optimises performance and security; usually holds certifications such as CCNA/CCNP.

**Goals and applications of networks (outline week 1)**

**Goals**: resource sharing (printers, files, Internet), communication (email, video), reliability (copies, alternative paths), cost reduction, centralised management, scalability, remote access. **Applications**: e-learning, e-banking, e-commerce, cloud services, IoT, telemedicine. **Disadvantages**: expensive to install, needs admin time, server failure can halt work, cables break, security costs.

Admin = **runs** it; engineer = **designs and builds** it. The hacker part (Q4e.iii) is in Pack 08.

## 9 Practice questions

**PRACTICE Q1 EASY** **[4 marks]**

State the four characteristics that determine the effectiveness of a data communication system.

**Answer**

**Delivery** (to the correct destination), **Accuracy** (no errors), **Timeliness** (on time; real-time for audio/video), **Jitter** (variation in packet arrival time must be small).

**PRACTICE Q2 EASY** **[3 marks]**

A company wants a full mesh between its 6 branch routers. How many links are needed and how many ports does each router need?

**Answer**

**Working**

$$\text{links} = \frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15$$

$$\text{ports} = n - 1 = 5 \text{ per router}$$ ← *one link to each other router*

**PRACTICE Q3 EASY** [3 marks]

Classify each as simplex, half duplex or full duplex: (a) a police two-way radio (b) a WhatsApp video call (c) a car radio receiving FM.

**Answer**

(a) Half duplex (b) Full duplex (c) Simplex.

**PRACTICE Q4 MEDIUM** [4 marks]

Distinguish between de facto and de jure standards and name two standards organisations.

**Answer**

**De facto** ("by fact"): adopted through widespread use and convention without approval by an official body, e.g. TCP/IP in its early days. **De jure** ("by law"): legislated and approved by an officially recognised body, e.g. IEEE 802.3. Organisations: ISO, ITU-T, ANSI, IEEE, EIA, IETF.

**PRACTICE Q5 MEDIUM** [5 marks]

Explain the three elements of a protocol, using an email message as the example.

**Answer**

**Syntax**: the format: an email has header fields (From, To, Subject) followed by a body, in a fixed order. **Semantics**: what the fields mean and the action: "To" tells the mail server where to deliver; a 550 reply means the mailbox does not exist, so the sender gives up. **Timing**: when and how fast: the client waits for the server's "250 OK" before sending the next command, so it does not overwhelm the server.

**PRACTICE Q6 MEDIUM** [6 marks]

Compare star and mesh topologies under the headings reliability, cost, installation and typical use.

**Answer**

| | Star | Mesh |
|---|---|---|
| **Reliability** | One cable fault affects one node, but the central switch is a single point of failure | Very high: many alternative paths, no single point of failure |
| **Cost** | Moderate: one cable per node + a switch | Very high: $n(n-1)/2$ links and $n-1$ ports per node |
| **Installation** | Easy to install, add and remove nodes | Difficult; adding a node means links to all others |
| **Typical use** | Office and lab LANs | WAN backbones between routers |

**PRACTICE Q7 MEDIUM** [4 marks]

A school lab uses Ethernet with all PCs cabled to a hub. State its physical and logical topology and explain why they differ.

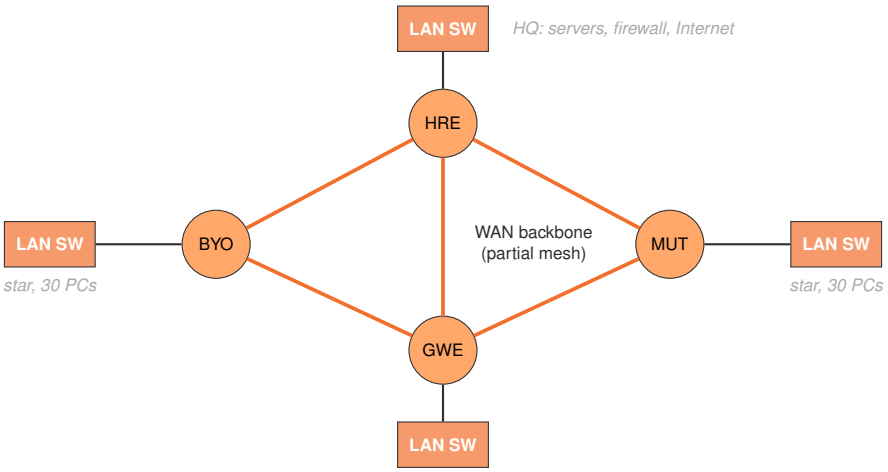
**Answer**

Physical: **star** (each PC has its own cable to the hub). Logical: **bus**, because a hub repeats every frame out of all ports, so every PC receives every frame and they share one collision domain, exactly as on a shared bus cable. Physical = how it is wired; logical = how the data flows.

**PRACTICE Q8 HARD** **[12 marks]**

A retail company has a head office in Harare and three branches (Bulawayo, Mutare, Gweru), each with about 30 staff. With the aid of a diagram, propose a design using a backbone network and local-access networks. Justify the topology and devices chosen at each level.

**Answer**




- **Backbone**: one router per site joined by leased lines / MPLS / fibre from a carrier, in a **partial mesh** so each site has at least two paths; if one link fails traffic reroutes. Full mesh (6 links) would cost more for little gain.
- **Local access**: in each site a **star** of PCs on a 48-port switch (or two 24-port switches, extended star), Cat6 UTP, a wireless AP; the switch up-links to the site router.
- **HQ**: servers (ERP, email, file, DHCP/DNS), firewall and the single Internet link; branches reach them over the backbone.
- **Addressing**: one subnet per site, e.g. 10.1.0.0/24, 10.2.0.0/24 . . .
- **Justification**: star is cheap, easy to manage, a fault affects one PC; the mesh backbone gives reliability where an outage would stop all trading; the two-level design scales by adding a site router and one more star.

**PRACTICE Q9 HARD** **[10 marks]**

Discuss the factors a network designer should consider when choosing a topology for a new hospital network, and recommend a topology.

**Answer**

1. **Reliability** / **fault tolerance**: patient systems must not stop; avoid bus and ring, where one break stops all.
2. **Cost**: budget for cable, switches and ports (mesh is costly).
3. **Cable length and building layout**: several wards and floors need a hierarchy.
4. **Future growth**: new wards must be added without disrupting others; star/tree expand easily.
5. **Ease of troubleshooting**: faults must be isolated quickly.
6. **Performance**: imaging (X-ray) files need high bandwidth, no shared collision domain.
7. **Security**: separate clinical, admin and guest traffic (VLANs).

**Recommendation**: a **hybrid**: an extended star (tree) in each building for local access, with redundant core switches joined in a **partial mesh** backbone, so no single device failure stops the hospital. Draw the core mesh with stars hanging off it.

**PRACTICE Q10 HARD** **[10 marks]**

“Peer-to-peer networks are obsolete; every organisation should use client-server.” Discuss this statement, with examples.

**Answer**

**For client-server**: centralised security and authentication; central backup and data integrity; easier management of users and updates; dedicated servers give consistent performance; essential for banks, universities (HIT e-learning), email and web services.
**Against (P2P still useful)**: cheap, no server or network OS to buy; quick for a small office of 2–10 PCs sharing a printer (Windows workgroup); no single point of failure; large-scale P2P (BitTorrent, blockchain) scales because every peer adds capacity and bandwidth; reduces load on central servers (e.g. software updates delivered peer to peer).
**Conclusion**: the statement is too strong. Client-server is best where security, central data and many users matter; P2P remains the right choice for very small networks and for distributed content sharing. Many systems are hybrid (a central index server + peer transfers).

**PRACTICE Q11 HARD** **[12 marks]**

With the aid of diagrams, explain the bus, ring and star topologies, and show what happens in each when one cable segment fails.

**Answer**

Draw each topology (Section 4) and mark a cross on one link.

- **Bus**: all nodes on one terminated backbone; a signal goes both ways and every node checks the address. **Backbone break**: the cable is no longer terminated, signals reflect and the **whole segment goes down**. A drop-cable break isolates only that node.
- **Ring**: each node connects to two neighbours; data travels one way and each node repeats it (token passing). **One break**: the loop is open, the token cannot circulate and the **whole ring stops** (a dual ring such as FDDI wraps round the fault).
- **Star**: each node has its own link to a central switch. **One cable break**: only that node is cut off; the rest work. But if the **switch** fails, everything stops.

Conclusion: star gives the best fault isolation for a LAN, which is why modern Ethernet LANs are stars.

## ONE-PAGE CHEAT SHEET

**5 components** (M-S-R-T-P)

- Message, Sender, Receiver, Transmission medium, Protocol
- 4 characteristics: Delivery, Accuracy, Timeliness, Jitter

**Transmission modes**

- Simplex: one way (keyboard, TV broadcast)
- Half duplex: both ways, one at a time (walkie-talkie)
- Full duplex: both ways at once (phone call)
- Point-to-point: dedicated link, 2 devices; multipoint: 3+ share a link

**Network sizes**

- PAN $\sim$10 m (Bluetooth); LAN building/campus, private, fast, cheap
- MAN city 5–50 km; WAN country/world, carriers, slower, costly
- Internet = network of networks (TCP/IP); WWW = web pages on it (HTTP)

**Protocol** = rules: what, how, when. Syntax, Semantics, Timing. HTTP, SMTP, TCP, IP, DNS, DHCP, FTP.
**Standards**: de facto (by use) vs de jure (by law). ISO, ITU-T, ANSI, IEEE, IETF.

**Topologies**

- **Bus**: one backbone + terminators; cheap; break = all down
- **Star**: central switch; easy, fault isolation; switch = single point of failure
- **Ring**: loop, one way, token; no collisions; one break stops ring
- **Mesh**: $n(n-1)/2$ links, $n-1$ ports; reliable, very costly
- **Tree**: hierarchy of stars on a root; scalable; root failure cuts segments
- **Hybrid**: mix (star-bus); flexible, complex
- Physical = how it is wired; logical = how data flows (hub: star wired, bus logic)

**Design**: backbone = fast core, mesh/partial mesh; local access = star/tree to users. 5 floors $\times$ 20 PCs: extended star, 24-port SW per floor, core SW, router, firewall, VLAN per floor.
**Client-server**: central server, security, backup, management, scalable; costly, server = SPOF. Web, email, HIT portal.
**P2P**: equal peers, cheap, no SPOF; weak security, scattered data. BitTorrent, workgroup.
**Admin** runs the network; **engineer** designs and builds it.